$MgBr_2(S)\ \Delta H_f^0 = -524\ KJ\ mol^{-1}$

Sublimation :

$Mg(S) \rightarrow Mg(g)$

$\Delta H_1^0 = +148\ KJ\ mol^{-1}$

Ionisation :

$Mg(g) \rightarrow Mg^{2+}(g) + 2e^-$

$\Delta H_2^0 = 2188\ KJ\ mol^{-1}$

Vapourisation :

$Br_2(1) \rightarrow Br_2(g)$

$\Delta H_3^0 = +31\ KJ\ mol^{-1}$

Dissociation :

$Br_2(g) \rightarrow 2Br(g)$

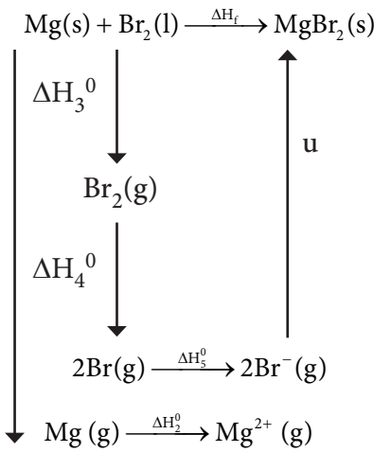
$\Delta H_4^0 = +193\ KJ\ mol^{-1}$

Electron affinity :

$Br(g) + e^- \rightarrow Br^-(g)$

$\Delta H_5^0 = -324.5\ KJ\ mol^{-1}$

Solution :



$\Delta H_f = \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4 + 2\Delta H_5 + u$

$-524 = 148 + 2188 + 31 + 193 + (2 \times -324.5) + u$

$-524 = 1911 + u$

$u = -524 - 1911$

$u = -2435\ kJ\ mol^{-1}$

5.

Solution:

Given

$T_h = 127^0\ C = 127 + 273 = 400\ K$

$T_C = 47^0\ C = 47 + 273 = 320\ K$

% efficiency η = ?

$$\eta = \left[\frac{T_h - T_C}{T_h}\right] \times 100$$

$$\eta = \left[\frac{400 - 320}{400}\right] \times 100$$

$$\eta = \left[\frac{80}{400}\right] \times 100$$

$\eta = 20\%$

6.

Solution:

Given:

$S^0\ (urea) = 173.8\ J\ mol^{-1}\ K^{-1}$

$S^0\ (H_2O) = 70\ J\ mol^{-1}\ K^{-1}$

$S^0\ (CO_2) = 213.5\ J\ mol^{-1}\ K^{-1}$

$S^0\ (NH_3) = 192.5\ J\ mol^{-1}\ K^{-1}$

$NH_2 - CO - NH_2 + H_2O \rightarrow 2NH_3 + CO_2$

$\Delta S_r^0 = \Sigma\ (S^0)_{products} - \Sigma\ (S^0)_{reactants}$

$\Delta S_r^0 = [2\ S^0\ (NH_3) + S^0\ (CO_2)]$

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